FORMULARIO DE PROCESAMIENTO DIGITAL DE SEÑALES (PDS)

Notación: $t \in \mathbb{R}$ [s], $n \in \mathbb{Z}$ [muestras], f [Hz], $\omega = 2\pi f$ [rad/s], $F = f/f_s$ [ciclos/muestra], $\Omega = 2\pi F$ [rad/muestra]

1. Muestreo, Frecuencias y Reconstrucción

$$x[n] = x_a(nT_s) = x_a\left(\frac{n}{f_s}\right), \quad T_s = \frac{1}{f_s}$$

$$F = \frac{f}{f_s} \in \left[-\frac{1}{2}, \frac{1}{2}\right], \quad \Omega = \omega T_s = 2\pi\frac{f}{f_s} = 2\pi F \in [-\pi, \pi]$$

$$f_{\text{Nyquist}} = 2f_{\text{máx}}, \quad f_s \geq 2f_{\text{máx}}, \quad f_N = \frac{f_s}{2} \iff \Omega_N = \pi$$

$$f_{\text{guarda}} = f_N - f_{\text{máx}} = \frac{f_s}{2} - f_{\text{máx}}$$

Interpolación Ideal (Sinc):

$$x_a(t) = \sum_{n=-\infty}^{\infty} x[n]\frac{\sin\left(\pi f_s(t - nT_s)\right)}{\pi f_s(t - nT_s)}$$

2. Cuantificación Uniforme PCM (B bits)

$$L = 2^B, \quad \Delta = \frac{V_{\text{máx}} - V_{\text{mín}}}{2^B - 1} \approx \frac{V_{\text{pp}}}{2^B} = \frac{2X_m}{2^B}$$

$$e_q[n] = x_q[n] - x[n] \in \left[-\frac{\Delta}{2}, \frac{\Delta}{2}\right], \quad e_{q,\text{máx}} = \frac{\Delta}{2}$$

$$\mu_e = 0, \quad \sigma_e^2 = \int_{-\Delta/2}^{\Delta/2} \frac{e^2}{\Delta}\, de = \frac{\Delta^2}{12}, \quad \sigma_e = \frac{\Delta}{\sqrt{12}}$$

Deducción de SNR$_Q$:

$$\sigma_e^2 = \frac{(2X_m/2^B)^2}{12} = \frac{X_m^2\, 2^{-2B}}{3}$$

$$\text{SNR}_Q = 10\log_{10}\left(\frac{\sigma_x^2}{\sigma_e^2}\right) = 10\log_{10}\left(3 \cdot 2^{2B}\frac{\sigma_x^2}{X_m^2}\right)$$

$$\text{SNR}_Q = 6{,}02B + 4{,}77 - 20\log_{10}\left(\frac{X_m}{\sigma_x}\right) \text{ [dB]}$$

Caso Senoidal ($x[n] = X_m\sin(\Omega_0 n) \implies \sigma_x = X_m/\sqrt{2}$):

$$\text{SNR}_{Q,\text{máx}} = 6{,}02B + 4{,}77 - 3{,}01 = \mathbf{6{,}02B + 1{,}76} \text{ [dB]}$$

3. Secuencias Elementales, Periodicidad y Energía

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases} \quad u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases} = \sum_{k=-\infty}^{n} \delta[k]$$

$$\text{Periodicidad: } \frac{\Omega_0}{2\pi} = F_0 = \frac{k}{N} \in \mathbb{Q} \implies N = \frac{2\pi k}{\Omega_0} \quad (k, N \in \mathbb{Z}^+)$$

$$x_e[n] = \frac{1}{2}\left(x[n] + x^*[-n]\right), \quad x_o[n] = \frac{1}{2}\left(x[n] - x^*[-n]\right)$$

$$E_x = \sum_{n=-\infty}^{\infty} |x[n]|^2, \quad P_x = \lim_{N\to\infty}\frac{1}{2N+1}\sum_{n=-N}^{N}|x[n]|^2$$

4. Serie de Fourier en Tiempo Continuo ($\omega_0 = 2\pi/T$)

$$x_a(t) = \frac{a_0}{2} + \sum_{k=1}^{\infty}\left[a_k\cos(k\omega_0 t) + b_k\sin(k\omega_0 t)\right]$$

$$a_0 = \frac{2}{T}\int_{-T/2}^{T/2} x_a(t)\, dt, \quad a_k = \frac{2}{T}\int_{-T/2}^{T/2} x_a(t)\cos(k\omega_0 t)\, dt$$

$$b_k = \frac{2}{T}\int_{-T/2}^{T/2} x_a(t)\sin(k\omega_0 t)\, dt$$

$$\text{Par: } b_k = 0,\ a_k = \frac{4}{T}\int_0^{T/2} x_a(t)\cos(k\omega_0 t)\, dt$$

$$\text{Impar: } a_0 = a_k = 0,\ b_k = \frac{4}{T}\int_0^{T/2} x_a(t)\sin(k\omega_0 t)\, dt$$

$$x_a(t) = \sum_{k=-\infty}^{\infty} C_k e^{jk\omega_0 t}, \quad C_k = \frac{1}{T}\int_{-T/2}^{T/2} x_a(t)e^{-jk\omega_0 t}\, dt$$

$$a_0 = 2C_0, \quad a_k = C_k + C_{-k}, \quad b_k = j(C_k - C_{-k}), \quad C_{\pm k} = \frac{a_k \mp jb_k}{2}$$

5. Transformada Z Bilateral y Pares Fundamentales

$$X(z) = \mathcal{Z}\{x[n]\} = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

Secuencia $x[n]$	**Transformada $X(z)$**	**ROC**
$\delta[n]$	1	Todo z
$\delta[n-n_0]$	z^{-n_0}	$z \neq 0$ ($n_0 > 0$); $z \neq \infty$ ($n_0 < 0$)
$u[n]$	$\frac{1}{1-z^{-1}} = \frac{z}{z-1}$	$\lvert z\rvert > 1$
$a^n u[n]$	$\frac{1}{1-az^{-1}} = \frac{z}{z-a}$	$\lvert z\rvert > \lvert a\rvert$
$-a^n u[-n-1]$	$\frac{1}{1-az^{-1}} = \frac{z}{z-a}$	$\lvert z\rvert < \lvert a\rvert$
$na^n u[n]$	$\frac{az^{-1}}{(1-az^{-1})^2} = \frac{az}{(z-a)^2}$	$\lvert z\rvert > \lvert a\rvert$
$-na^n u[-n-1]$	$\frac{az^{-1}}{(1-az^{-1})^2} = \frac{az}{(z-a)^2}$	$\lvert z\rvert < \lvert a\rvert$
$\cos(\Omega_0 n)u[n]$	$\frac{1-\cos(\Omega_0)z^{-1}}{1-2\cos(\Omega_0)z^{-1}+z^{-2}}$	$\lvert z\rvert > 1$
$\sin(\Omega_0 n)u[n]$	$\frac{\sin(\Omega_0)z^{-1}}{1-2\cos(\Omega_0)z^{-1}+z^{-2}}$	$\lvert z\rvert > 1$
$r^n\cos(\Omega_0 n)u[n]$	$\frac{1-r\cos(\Omega_0)z^{-1}}{1-2r\cos(\Omega_0)z^{-1}+r^2z^{-2}}$	$\lvert z\rvert > r$
$r^n\sin(\Omega_0 n)u[n]$	$\frac{r\sin(\Omega_0)z^{-1}}{1-2r\cos(\Omega_0)z^{-1}+r^2z^{-2}}$	$\lvert z\rvert > r$

6. Propiedades de la Transformada Z

Propiedad	**Tiempo**	**Dominio z**	**ROC**
Linealidad	$ax_1[n] + bx_2[n]$	$aX_1(z) + bX_2(z)$	$\supseteq \text{ROC}_1 \cap \text{ROC}_2$
Desplazamiento	$x[n-n_0]$	$z^{-n_0}X(z)$	ROC_x ($\pm 0, \infty$)
Escalamiento	$z_0^n x[n]$	$X(z/z_0)$	$\lvert z_0\rvert \cdot \text{ROC}_x$
Inversión	$x[-n]$	$X(z^{-1})$	$1/\text{ROC}_x$
Conjugación	$x^*[n]$	$X^*(z^*)$	ROC_x
Derivación	$nx[n]$	$-z\frac{dX(z)}{dz}$	ROC_x
Convolución	$x_1[n] * x_2[n]$	$X_1(z)X_2(z)$	$\supseteq \text{ROC}_1 \cap \text{ROC}_2$

$$\text{T. Valor Inicial ($x[n]$ causal): } x[0] = \lim_{z\to\infty} X(z)$$

$$\text{T. Valor Final: } \lim_{n\to\infty} x[n] = \lim_{z\to 1}(1 - z^{-1})X(z)$$

7. Sistemas LIT, Función de Transferencia y Estabilidad

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

$$y[n] + \sum_{k=1}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k x[n-k]$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^{M} b_k z^{-k}}{1 + \sum_{k=1}^{N} a_k z^{-k}} = z^{N-M}\frac{\sum_{k=0}^{M} b_k z^{M-k}}{z^N + \sum_{k=1}^{N} a_k z^{N-k}}$$

- **Causalidad:** $h[n] = 0$ $(\forall n < 0) \iff$ ROC: $|z| > |p_{\text{máx}}|$.
- **Estabilidad BIBO:** $\sum_{n=-\infty}^{\infty}|h[n]| < \infty \iff \{|z| = 1\} \subset \text{ROC}$.
- **Causal y BIBO Estable:** $|p_k| < 1 \quad (\forall k = 1, \ldots, N)$.

8. Estructuras de Realización: Forma Directa I y II

Parámetro	**Forma Directa I**	**Forma Directa II (Canónica)**
Retardos z^{-1}	$M + N$	$\max(M, N)$
Nodo interno	$v[n] = \sum_{k=0}^{M} b_k x[n-k]$	$w[n] = x[n] - \sum_{k=1}^{N} a_k w[n-k]$
Salida $y[n]$	$y[n] = v[n] - \sum_{k=1}^{N} a_k y[n-k]$	$y[n] = \sum_{k=0}^{M} b_k w[n-k]$

Relación con ganancias de lazos de realimentación ($g_k = -a_k$):

$$w[n] = x[n] + g_1 w[n-1] + g_2 w[n-2] + \cdots + g_N w[n-N]$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{b_0 + b_1 z^{-1} + \cdots + b_M z^{-M}}{1 - g_1 z^{-1} - g_2 z^{-2} - \cdots - g_N z^{-N}}$$

9. Transformadas Frecuenciales: DTFT, DFT y FFT

$$\text{DTFT: } X(e^{j\Omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\Omega n} = X(z)\Big|_{z=e^{j\Omega}}$$

$$\text{IDTFT: } x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\Omega})e^{j\Omega n}\, d\Omega$$

$$\text{DFT } (W_N = e^{-j2\pi/N}): X[k] = \sum_{n=0}^{N-1} x[n]W_N^{kn}, \quad 0 \leq k \leq N-1$$

$$\text{IDFT: } x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k]W_N^{-kn}, \quad 0 \leq n \leq N-1$$

Mariposa FFT Radix-2 (Cooley-Tukey):

$$X[k] = E[k] + W_N^k O[k], \quad X\left[k + \frac{N}{2}\right] = E[k] - W_N^k O[k]$$

Costo DFT: $\mathcal{O}(N^2)$ | Costo FFT: $\mathcal{O}(N \log_2 N)$

10. Filtros Digitales: FIR vs. IIR y Familias Clásicas

Atributo	**FIR (No Recursivo / MA)**	**IIR (Recursivo / ARMA)**
Duración $h[n]$	Finita (M muestras)	Infinita ($n \to \infty$)
Polos en z	Origen ($z = 0$)	En cualquier $p_k \neq 0$
Estabilidad	Siempre estable	Estable si $\lvert p_k \rvert < 1$ (causal)
Fase	Lineal si $h[n] = \pm h[M-1-n]$	No lineal ($\tau_g(\Omega) \neq$ cte)
Orden req.	Alto (M grande)	Bajo (N reducido)

Familia IIR	**Banda de Paso**	**Banda de Rechazo**	**Transición / Orden**
Butterworth	Máx. plana	Monótona	Suave (mayor N)
Chebyshev I	Equirrizado	Monótona	Abrupta (N medio)
Chebyshev II	Monótona	Equirrizado	Abrupta (N medio)
Elíptico (Cauer)	Equirrizado	Equirrizado	Más abrupta (N mín.)

11. Transformada Bilineal, Fase Cero y Hilbert

$$\text{T. Bilineal: } s = \frac{2}{T_s}\left(\frac{1 - z^{-1}}{1 + z^{-1}}\right), \quad \omega_a = \frac{2}{T_s}\tan\left(\frac{\Omega_d}{2}\right) = \frac{2}{T_s}\tan\left(\frac{\omega_d T_s}{2}\right)$$

$$\text{Retardo Grupo: } \tau_g(\Omega) = -\frac{d\phi(\Omega)}{d\Omega}$$

Fase Cero (`filtfilt`): $v[n] = x[n] * h[n],\ g[n] = v[-n],\ r[n] = g[n] * h[n],\ y[n] = r[-n]$

$$Y(e^{j\Omega}) = X(e^{j\Omega})H(e^{j\Omega})H^*(e^{j\Omega}) = X(e^{j\Omega})|H(e^{j\Omega})|^2 \implies \phi_{\text{total}}(\Omega) = 0$$

$$\text{Señal Analítica: } z_a[n] = x[n] + j\hat{x}[n], \quad \hat{x}[n] = \mathcal{H}\{x[n]\}, \quad A[n] = \sqrt{x^2[n] + \hat{x}^2[n]}$$

12. Mapeo Plano s (Laplace) $\leftrightarrow$ Plano z y Eje Imaginario ($\mathrm{Re}(s) = 0$)

$$s = \sigma + j\omega \quad (\sigma = \mathrm{Re}\{s\},\ \omega = \mathrm{Im}\{s\}\ \text{[rad/s]}), \quad t = nT_s$$

$$z = e^{sT_s} = e^{(\sigma + j\omega)T_s} = e^{\sigma T_s} \cdot e^{j\omega T_s} = re^{j\Omega}$$

$$r = |z| = e^{\sigma T_s}, \quad \Omega = \omega T_s = 2\pi\frac{f}{f_s}\ \text{[rad/muestra]}, \quad \omega_s = \frac{2\pi}{T_s}$$

$\sigma = 0 \implies s = j\omega \implies r = |z| = e^{0 \cdot T_s} = 1 \implies z = e^{j\Omega}$ (Círculo Unitario)

$\sigma < 0 \implies |z| = e^{\sigma T_s} < 1$ (Semiplano Izq. $\mapsto$ Interior de $|z| = 1$)

$\sigma > 0 \implies |z| = e^{\sigma T_s} > 1$ (Semiplano Der. $\mapsto$ Exterior de $|z| = 1$)

CTFT: $X_a(j\omega) = X_a(s)\Big|_{s=j\omega} \iff$ DTFT: $X(e^{j\Omega}) = X(z)\Big|_{z=e^{j\Omega}}$

13. Identidades de Euler

$$e^{j\theta} = \cos(\theta) + j\sin(\theta), \quad e^{-j\theta} = \cos(\theta) - j\sin(\theta)$$

$$\cos(\theta) = \frac{e^{j\theta} + e^{-j\theta}}{2}, \quad \sin(\theta) = \frac{e^{j\theta} - e^{-j\theta}}{2j}, \quad \left|e^{j\theta}\right| = \sqrt{\cos^2(\theta) + \sin^2(\theta)} = 1$$